

Elevator People Counting

Designing an elevator people counting solution for the NuMicro® M55M1 series using the NuMaker M55M1

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Problem Statement

Problem

- Elevators lack accurate real-time passenger counts.
- Weight sensors cannot reliably determine occupancy.
- Leads to inefficient dispatching and potential overcrowding risks.



Solution

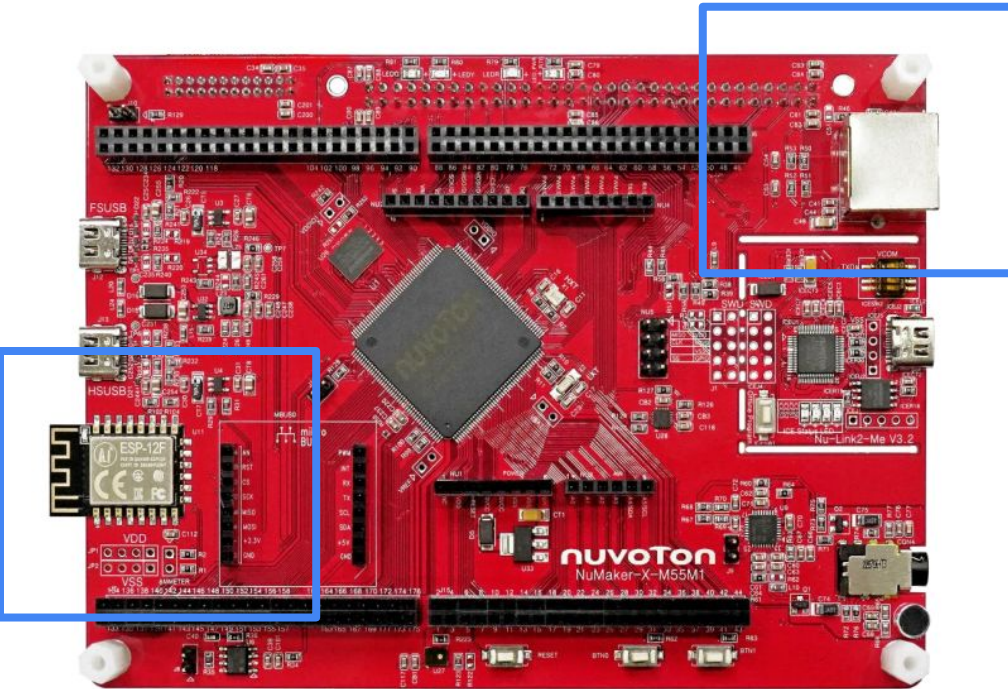
- Develop a privacy-preserving Edge AI system that counts people in real time.
- Image processing runs entirely on the Nuvoton NuMaker-X-M55M1D board.
- Achieve low-latency, high-accuracy occupancy monitoring.

NuMaker Details and Peripherals

Problem: How can we feed video to board, and extract passenger counts?



NuMaker Details



Ethernet PHY Module

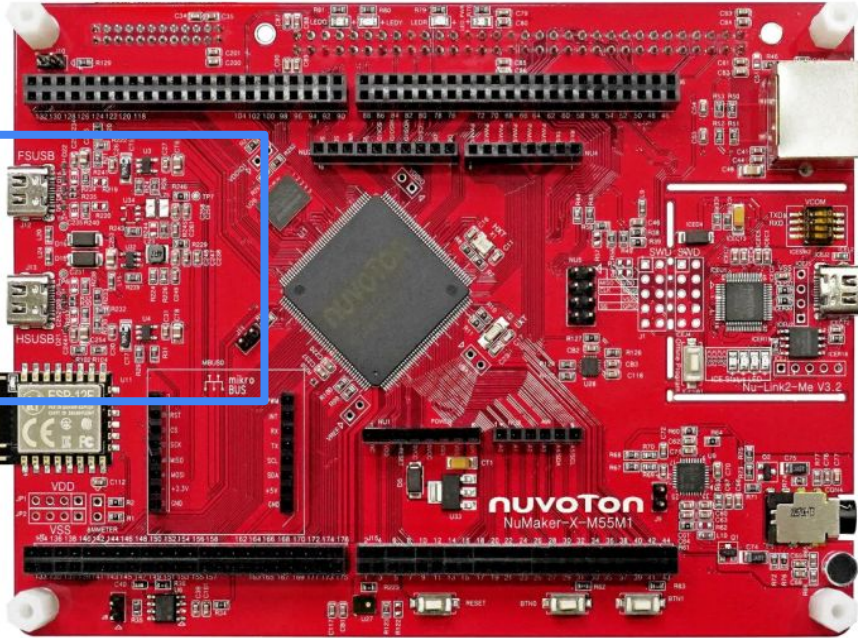
- Sustains 100 Mbps throughput
- Direct Memory Access (DMA) bridge to the SRAM

ESP Wi-Fi Module

- UART serial bus
- no DMA

55M1D_EN_Rev1.01.pdf

NuMaker Details



USB (HSUSB)

- DMA controller
- 480 Mbps

CCAP and Camera

- Peripheral DMA



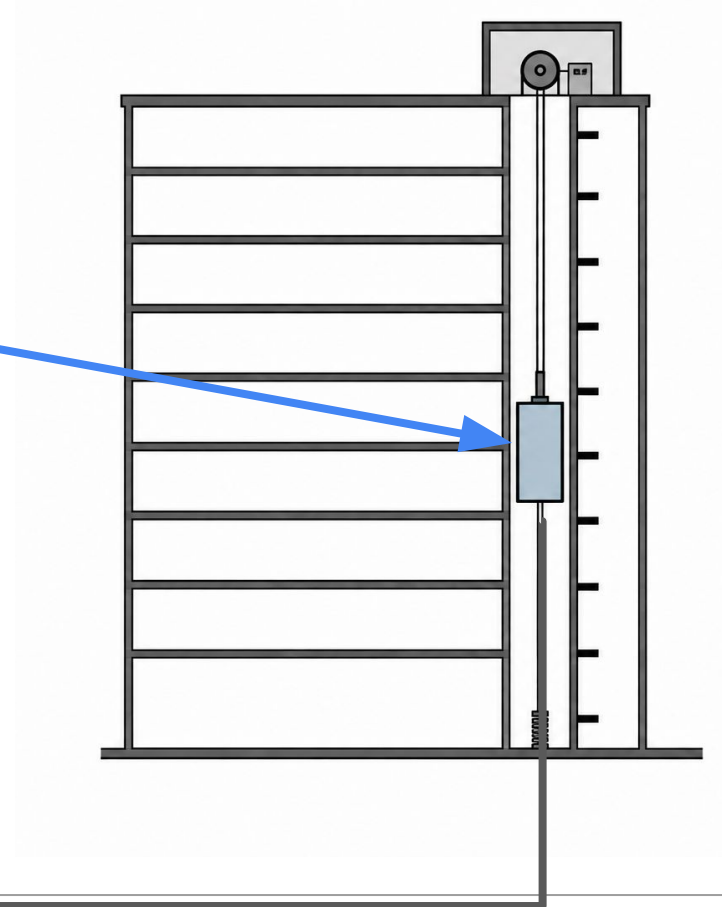
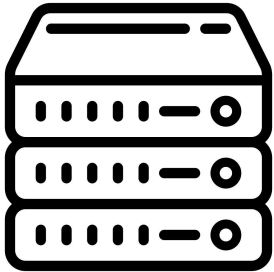
https://www.nuvoton.com/export/resource-files/en-us--UM_NuMaker-X-M55M1D_EN_Rev1.01.pdf

Approaches to Elevator Sensors

Elevator Installation (Theoretical)



passenger_count: 6

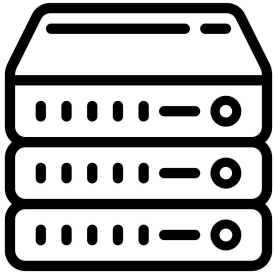


Serial line

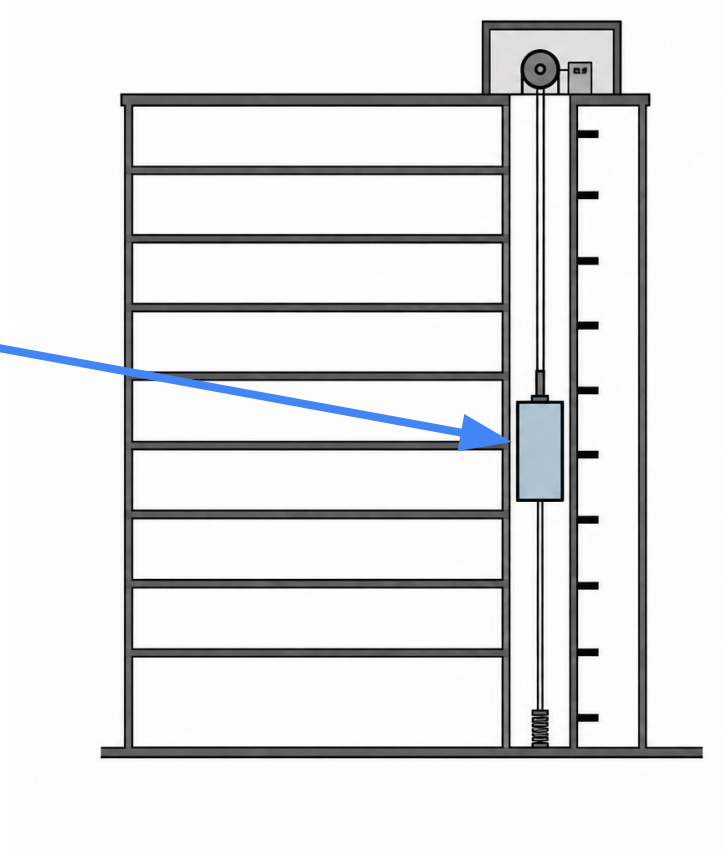
Elevator Installation (Theoretical)



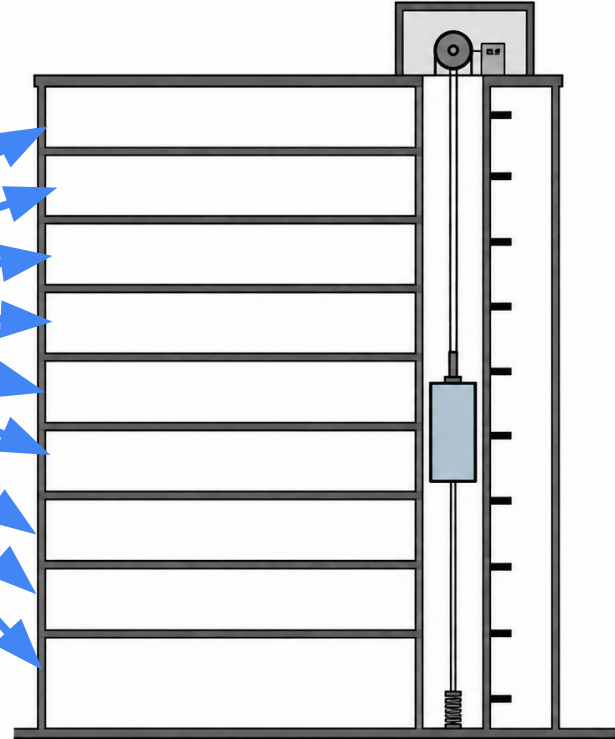
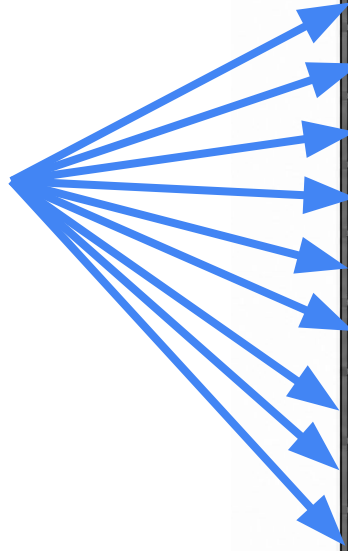
passenger_count: 6



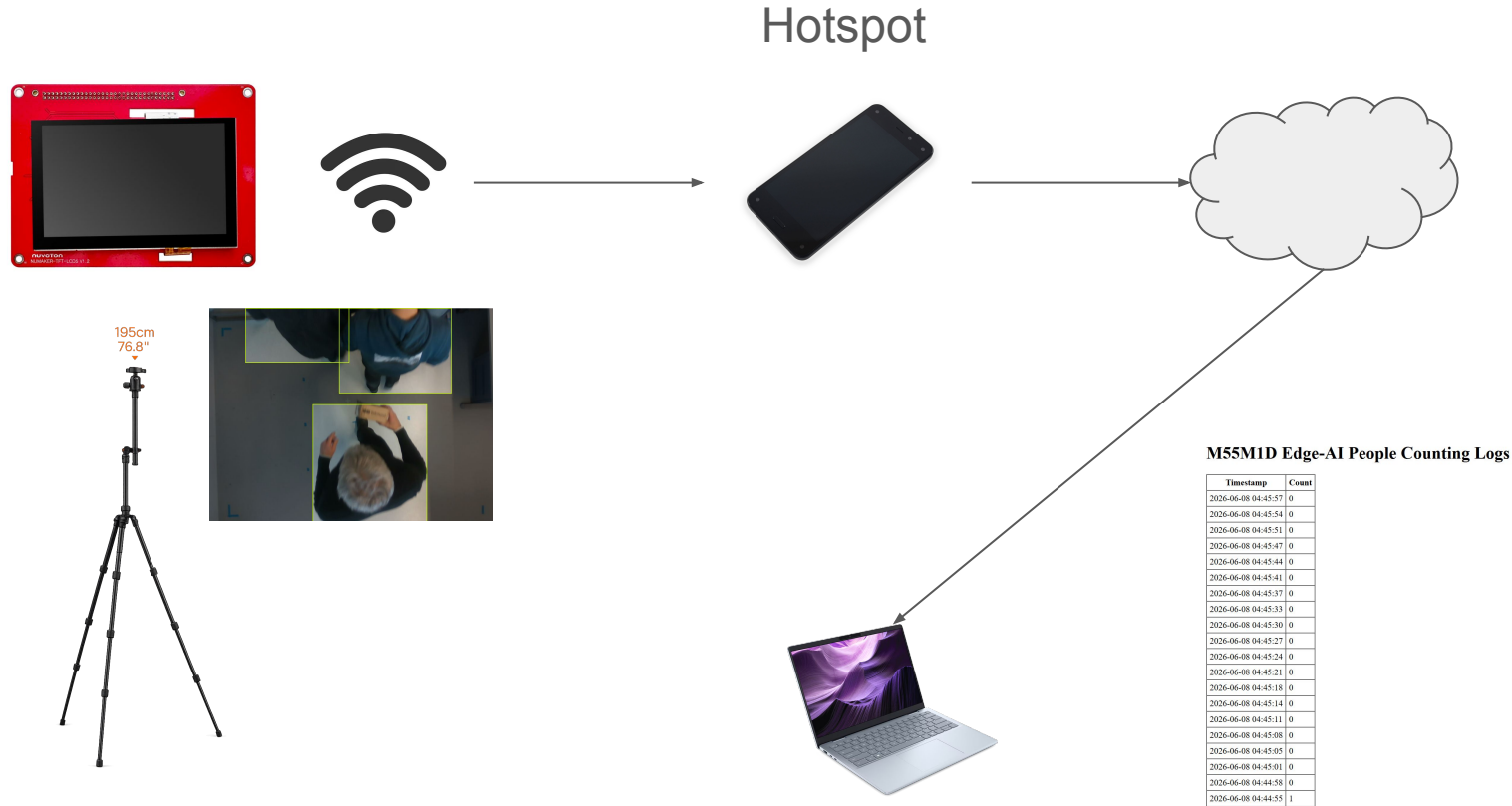
Wi-Fi



Floor-level Installation (Theoretical)



Approach



<https://www.amazon.com/dp/B0CRYWSKGM>

Collected Datasets

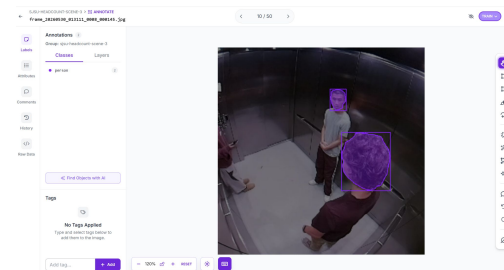
[Overhead-person-detection](#) (13.4k)



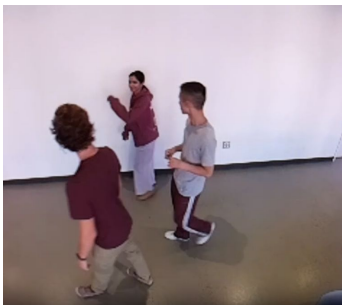
[overhead-people-rgb](#) (8.5k)



Roboflow + SAM3 Annotations



[sjsu-headcount-scene-1](#) (580)



[sjsu-headcount-scene-2](#) (350)

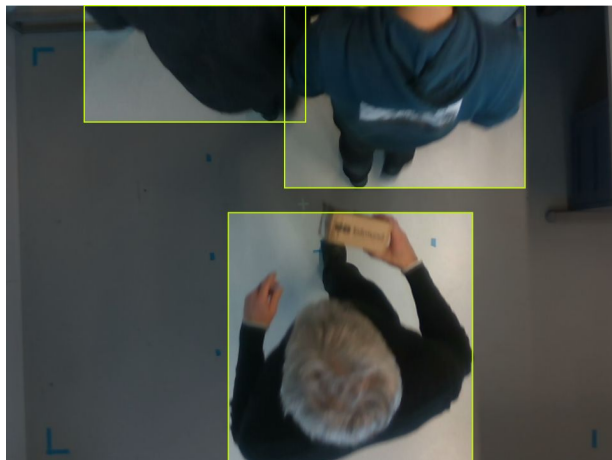


[sjsu-headcount-scene-3](#)



Computer Vision Models

Bounding Boxes Based (YOLOv8n-ReLU6)



Metric	Value
Precision	0.949
Recall	0.941
mAP@0.5	0.978
mAP@0.5:0.95	0.686

Point-Coordinate Based (FOMO)



Metric	Value
Precision	0.8108
Recall	0.8276
MAE	0.43

~80.73% overall test counting accuracy on scene-1

Final Recommendations

elevator people counting solution

YOLOv8n-ReLU6

Superior metrics, generalizes better and works out of the box in most cases

2 MB total (requires SD card)

Bounding-box based (CPU/energy intensive)

FOMO

Poor generalization, requires fine tuning to specific environments

64KB total (does not require SD Card)

Coordinate based (CPU efficient)

- We also recommend flexible camera FOVs (70/160/180) options to handle overhead views in different scenarios.
- Prepared casings for faster prototype installations onto elevators (case + magnet attachment)

Code and dataset directory

- <https://github.com/bencejdanko/Edge-AI-People-Counting-NuMaker-X-M55M1D>
 - YOLOv8n + Wi-Fi Connection + Log Dashboard

- <https://huggingface.co/datasets/bdanko/sjsu-people-counting>
 - SJSU collected video (split into scene-1, scene-2, scene-3)
 - <https://huggingface.co/datasets/bdanko/sjsu-headcount-scene-1>
 - <https://huggingface.co/datasets/bdanko/sjsu-headcount-scene-2> (Elevator)
 - <https://huggingface.co/datasets/bdanko/sjsu-headcount-scene-3> (Elevator)

Thank you Demo Q&A

Live demo logs at <https://elevator.visionindustry.com>